

UXL to develop open source AI accelerator software



The UXL Foundation, a consortium of big techs that aims for an open standard accelerator software ecosystem, now aims to challenge Nvidia's AI dominance.

The consortium plans to develop an open source suite of software and tools to power various AI accelerator chips, an area currently dominated by Nvidia's CUDA software platform.

The UXL's technical steering committee is set to finalise technical specifications in the first half of the year, with plans to mature the details by year-end. In the long-term, its goal is to attract a critical mass of developers to their platform, and the platform will eventually support Nvidia hardware and code also. But for now the aim is to break the monopoly.

The Linux Foundation announced the formation of the UXL Foundation in September, last year. It is an evolution of Intel's oneAPI initiative. OneAPI is an open standard programming model that helps developers to write software that can run efficiently for various types of hardware, including CPUs, GPUs, FPGAs, and accelerators.

The consortium consists of industry biggies including Arm, Fujitsu, Google Cloud, Imagination Technologies, Intel, Qualcomm Technologies, Inc., and Samsung. Along with chipmakers, UXL is also looking to add cloud-computing companies like Amazon and Microsoft's Azure.

This is one of the many efforts to reduce Nvidia's hold on the AI chip market through CUDA. Reportedly, venture financiers and corporate funds have poured more than \$4 billion into 93 separate efforts. The startups that aim to dethrone Nvidia received \$2 billion funding in 2023 up from \$580 million in 2022.

CUDA has been in use for 15 years now and there are plenty of developers who have built their code around it. So, in response to the rising effort to challenge its AI dominance, Nvidia's executive Ian Buck confidently said, "The world is getting accelerated. New ideas in accelerated computing are coming from all across the ecosystem, and that will help advance AI and the scope of what accelerated computing can achieve."

ONDC transforms Indian e-commerce, nears 50 million transactions

In a significant milestone for India's digital commerce landscape, the government-backed Open Network for Digital Commerce (ONDC) has facilitated an impressive 49.79 million transactions since its formal launch in January 2023. The mobility sector leads the charge in this digital revolution, with ride-hailing apps like Namma Yatri accounting for more than half of the total monthly orders.

BeagleY-AI: An SBC built to revolutionise AI development

BeagleY-AI is an open source single-board computer (SBC) developed by BeagleBoard.org, designed to revolutionise the creation of AI applications. It is engineered to optimise AI performance across various sectors, including robotics, factory and building automation, test and measurement, medical technology, human-machine interfaces (HMIs), and intelligent vision.

The board has a robust 64-bit, quad-core A53 processor, multiple AI accelerators alongside C7x DSPs, a 50 GFLOP GPU supporting up to three concurrent display outputs, and modern connectivity options including USB3.1, PCIe Gen 3, WiFi6, and BLE 5.4. It is also compatible with a wide array of existing accessories, such as Power over Ethernet (PoE), NVMe storage, and 5G connectivity.

Christine Long, CEO of BeagleBoard.org, expressed her enthusiasm: "The launch of BeagleY-AI as an open source board is a thrilling advancement in our mission to expand the open source hardware ecosystem. This platform not only fosters collaboration and learning but also empowers individuals and professionals to develop open designs and contribute to the broader community."

Romain Soreau, head of single board computing at element14, stated, "This is a significant development for all developers. BeagleY-AI offers an affordable and user-friendly platform, providing a seamless development experience with BeagleBoard's proven custom Debian Linux image."

The board is available for pre-order online through Farnell in EMEA, Element 14 in APAC, and Newark in North America, with shipping set to commence at the end of May.

In line with its commitment to fostering the growth of the open source hardware ecosystem, BeagleBoard.org has introduced six new boards over the past two years, aiming to make IoT and AI technologies more accessible.